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Course Name :- Natural language processing

Code :- DSA0308

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Continuous Internal Assessment - 1

1) A hospital information :-

Given information :-

Appointment on 05/03/2025 at 10:30

Call 9876543210 or +91-9123456789.

mail ravi.k@college.edu.in. Design a regular Expression

(a) a strict dd/mm/yyyy date (01-31, month 01-12)

~~\b(0~~ Give day (01-31), month (01-12)

~~\b(0-9)~~

\b[0[1-9][12][0-9]|3[01])/(0[1-9]|[0-2])/

[0-9]$4y\b

\b → word boundaries

0(1-9) → it matchs with 01 to 09 (days)

12(0-9) → it matchs with 10 to 29 (days)

3(01) → it matchs with 30, 31 (days)

0(1-9)|(0-2) → it matchs with 01 to 12 (months)

[0-9]{4y → it matchs with 01 for 12 (months)

\b → word boundaries.

(6) E-mail

(?<!\d)(?:\+91[-]?|0)?[6-9]\d

$93 (?!\d)

The mail Id should start with user name

the mail should contain numbers.

The mail have the symbol '@'.

L1 05/03/2025 valid

L2 9876543210 valid

L3 5876543210 invalid

L4 5/3/26 invalid

* L3 is invalid because the Starting number should be start between 6 & 9

* L4 is invalid because in With no zero

Exit.

E-

2) The lexicon contains: reg-noun = {fox, cat, dog};
irreg-sg-noun = {goose, mouse};
irreg-pl-noun = {geese, mice};
Plural suffix - s.

(a) Design a finite-state automaton for English nominal inflection.

The finite-state automato is the part of Natural language processing it is used to understand the Patterns from the text

They are 3 types

lexion level
intermidate level
Final level.

lexion level:-
It is original form of the word

Ex:- fox, cat, ~~dog~~ dog

intermidate level:-
It is the bansanformation of a word.

fox + es
cat + s

Final level
It is final output of word Ex:- foxes, cats

Using your automaton

Whether each of these morpheme sequences is accepted or not

(i) dog + -s = dogs

Hence dogs is accepted.

(ii) mouse + -s = mouses

Hence mouses is accepted

(iii) geese = geese

Hence it is accepted

(iv) fox - s = foxes foxs

(v) Hence foxs is not accepted

(v) mice + s = mices

Hence mices is accepted

stats	bansations
fox	Accepted
cat	accepted
dog	accepted
goose	accepted
mouse	rejected
geese	accepted
mice	rejected.

Morpheme sequence

stats	bansations.
dogs	Accepted
mouse	rejected → (Irregular noun)
geese	accepted
foxs	accepted
mices	rejected → Already Plural

Morphological:-

Dividing word into the Simpiler form.

Ex:- Walked - infectional, unhappiness - derivational

(3) (i) The porter stemmer (original algorithm:

Stip by stip words ponies, motoring, relational, hopefulness and generalizations.

It is used to divide the text into simplies understandable word form.

Steps :-

By removing the plural form of the word

By adding the Suffix of the word

By removing the additional suffix to the word.

By Analyzing the word

Generating the word.

[S_1 = "the cat sat"

S_2 = "the dog sat"

S_3 = "The dog ran")

Name	stips	Stages
Ponies	Poni + es	Poni
Relational	Relat + ional	relat
generalization	General + ization	genual.
hopefulness	hope + fulness	hope
motoring	motor + ing	motor.

b) N-gram model:-

The N-gram model is the predicting of the next word in the Sentence

they are two types:-

Unigram model:-

A Unigram model is the probability of the particular word.

bi-gram model:-

A bigram model is the probability of word depends on the front word.

S_1 = "the cat sat"

S_2 = "the dog sat"

S_3 = "the dog ran"

Entropy per word:-

Unigram

The = 3

cat = 1

dog = 2

Sat = 1

run = 1

(4) Small Corpus contains the three Sentences

<s> I like tea </s>,
<s> I like coffee </s>
<s> I drink tea </s>.

Given information

V = 6 (I, like, tea, coffee, drink)
N = 12 (Word tokens excluding <s>.

N-gram model:-

The N-gram model is the predicting of the Next word in the Sentences.

They are two types:-

Unigram model:-

A Unigram model is the probability of the Particular word.

bi-gram model:-

A bigram model is the probability of word depends on the front word.

ref

S_1 = I like tea
S_2 = I like coffee
S_3 = I drink tea.

Unigram :-

the = 1

Tea = 1

coffee = 2

[illegible] = 1

like = 1

drink = 2

(5) (a) Viterbi Algorithm

$V_1(PRP) = 0.5 \times 0.8$

$= 0.4$

V_2 for "can" vs for "fish"

MD $0.4 \times 0.4 \times 0.3$

$= 0.048$

VB $0.4 \times 0.5 \times 0.02$

$= 0.004$ max

$(0.048 \times 0.8, 0.004 \times 0.1, 0.0004 \times 0.4) \times 0.01$

$= 0.0384 \times 0.01$

$= 0.000384$

NN $0.4 \times 0.1 \times 0.01 = 0.0004$ max $(0.048 \times 0.2, 0.004 \times 0.5, 0.0004 \times 0.3) \times 0.02$

$= 0.0096 \times 0.02$

$= 0.000192.$

Best final state VB. Bank traning gives they/PRP can/MD fish/VB with 3.84×10^{-4} Probability.

Rule 1 :- If word is can with candidate tags and the precious word is a personal pronoun (PRP)

Eliminate NN & VD so can = MD

Rule 2 :- If the previous tag is MD, Eliminate NN

So fish = VB.

constraints rule progressively eliminate reading from the ambiguous set until one tag is left.

(c) Transformation Based

They/PPP can/MD fish/NN) which has 1 error out of 3 tokens. the learner tries template over the Previous tag. Context & Selects the best : namely "change NN to VD ~~with~~ when Score = fixes - breaks

= 46 - 6 They PRP can/MD fish/VB with 0 errors.
= 40

Comparison :-

Approach	knowledge	Strength	Weakness
Rule-based approach	Hand-written constraint rules	powerful, high precision	costly to write, hard to maintain.
Stochastic (HMM) Tag & word.	Probabilities from a corpus,	learns automatically ranks alternatives	Need a tagged corpus ; sparse data.